

1. Exposure

Areca nut and smokeless tobacco are held against the gingivobuccal mucosa.

2. Chronic injury

Sustained irritation of the buccal and gingival epithelium, for years.

3. Inflammation

A persistent inflammatory infiltrate in the subepithelial stroma.

8. Progression

Leukoplakia advances to invasive gingivobuccal carcinoma.

7. Effector programme

MMP9, PTGS2, CCND1, VEGFA and BCL2 sustain growth and invasion.

6. Chemokine output

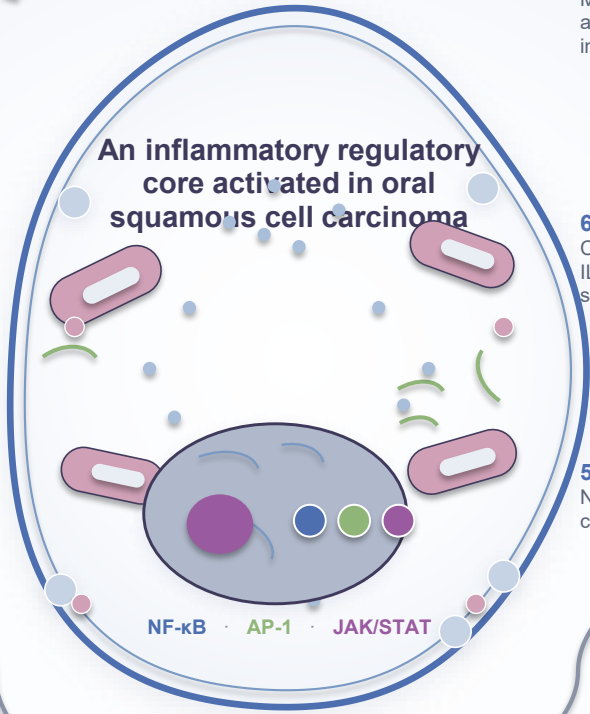
CXCL8, CXCL10, CCL5, IL6 and IL1B are released into the stroma.

5. Nuclear action

NFKB1 mRNA is unchanged: the core is activated after translation.

4. Regulatory core

NF- κ B, AP-1 and JAK/STAT transcription factors dominate the causal network.



Network-nominated

A dependency network blind to oral cancer.



Cohort replication

Up in two Indian cohorts: 61 tumours, 20 controls.



Not an artefact

Survives immune and stromal adjustment.



Core, not edges

Genes replicate in TCGA-HNSC; edges do not.